



Quantitative Structural Health Monitoring of Composite Materials

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Abstract. Fibre Reinforced Composites (FRCs) are extensively used for manufacturing high-value structural engineering components. The complex damage mechanisms affecting such materials and the difficulty in inspecting them with traditional non-destructive testing (NDT) methods increases the need for effective Structural Health Monitoring (SHM) systems. SHM can evaluate FRC components in real time while they are in service, reducing downtime and minimising the risk of unexpected catastrophic failure. In this study, we report on the development of a customised acoustic emission (AE) system which is capable of capturing the full waveform and subsequently analyse it using various algorithms. Through this novel approach the identification, characterisation and quantification of damage initiation and evolution has been assessed on FRC coupon samples during tensile and flexural testing. Moving RMS has been one of the algorithms exploited for identification and quantification of damage during loading to failure of the tested FRC coupon samples. From the results obtained it has been possible to detect and quantify damage initiation and evolution through the identification of key events, including delamination, in the samples tested.

1 Introduction

Fibre Reinforced Composites (FRCs) combine light weight with high strength and stiffness, and high corrosion resistance (1). They are the materials of choice in manufacturing critical aircraft components, wind and tidal turbine blades, structural components for high-performance and military land vehicles, and high-end sporting equipment such as bicycles and skis. The advances made in FRC manufacturing technology have allowed wind turbines to increase the rotor size by several times, reaching blade lengths up to 131 m (2). This has enabled the largest offshore wind turbines to achieve power ratings that exceed 14 MW (3). However, this means that failure of Wind Turbine Blades (WTBs) can not only result in loss of electricity production, due to unnecessary down-time, but also damage the wind turbine beyond repair. Therefore, accurate remote condition monitoring of the structural integrity of WTBs and other critical structural components made of FRCs has an important role to play. Acoustic Emission (AE) has been extensively used as a means of qualitatively assessing the structural integrity of FRCs in-service. Several published studies have looked into the analysis of AE signals in order to identify damage modes occurring in such materials, including matrix cracking, fibre breakage, fibre pull-out, debonding, and delamination (4–6). However, the majority of such studies have achieved qualitative or semi-quantitative evaluation of AE datasets, being unable to provide accurate quantitative results. In the present paper we discuss a novel approach for the quantitative evaluation of damage in FRCs based on mechanical tests carried out on coupon samples in conjunction with AE monitoring under laboratory conditions.

2 Methodology

The datasets used for analysis within this paper were produced from the monitoring of mechanical testing

completed within the scope of H2020 Carbo4Power project, at the University of Birmingham (UoB). The overreaching aim of this project is the improvement of materials and technologies associated with the production and operation of wind turbines, with specific focus on our part being WTBs. Association with this meant that no singular material or testing mode was subject to particular attention, and that due to the nature of construction of WTBs both Carbon Fibre Reinforced Polymers (CFRP) and Glass Fibre Reinforced Polymers (GFRP) were held in consideration and tested. It should be stated that unless specified otherwise, the findings presented hold true for both of these materials.

2.1 Acoustic Emission Monitoring

For the AE monitoring of the materials two separate signal acquisition systems were utilised, a commercially available system, from Physical Acoustics Corporation (PAC), and a customised system developed in house at UoB. The customised system operates off a custom Data Acquisition (DAQ) software developed at UoB. For connectivity between the hardware and the software a National Instruments NI-9223 DAQ, capable of capturing simultaneously 4 channels at 1 MS/s within the range of +/- 10V was utilised.

2.2 Materials Tested

All sample coupons were manufactured in accordance with ASTM standards. Tensile test coupons were of dimensions 250 mm x 25 mm and of thickness ranging from 1 to 2 mm; with this being dependent on the reinforcement type. For flexural testing, the test coupons had dimensions of 150 mm x 25 mm and a thickness of ~1.3 mm. The test span, the distance between the two securing points in the three-point bend test, was of 40 mm.

It is important to note that these tests were not completed with the aim of material certification, but for

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their instrumentation for AE monitoring. With this reasoning, little consideration is required for the mechanical properties, as the focus is simply on the identification of damage occurrence.

2.3 Mechanical Testing

Two types of mechanical testing were utilised including tensile testing and flexural (3-point bending) testing. Tensile testing was completed using a Zwick Roell, model 1484, electro-mechanical testing machine a 200 kN load cell, 10 mm extensometer length and a crosshead movement of 2 mm/min. The three-point bending was completed in the Dartec, using a 50 kN load cell instead of a 1 kN load cell due to lack of availability, with a crosshead movement speed of 1 mm/min.

3 Results

It should be noted that it is not feasible to filter all mechanical noise through its application, as due to the nature of the testing, this will vary heavily throughout the test, with noise of greater amplitude overlapping with those signals associated with the occurrence of damage. All that can be expected is to limit the impact of noise on the AE signal and its discernible information. The sensor type being utilised is designed to assist in limiting the interference caused by mechanical noise in the acquisition of an AE signal.

The upper and lower threshold values can be discerned, in this case, through the evaluation of the early sections of the signal, e.g. the initial sample of 5 seconds. Loading in this region will not be significant enough to cause the onset of damage, such that the mechanical noise that is associated with the operation of the tensile test machine can be assumed to be the primary source of AE peaks observed. In essence the isolation of this will serve to act as an estimation of the activity associated with the general operation of the machinery. This should not be confused with standardised threshold to acquisition, what is observed within commercial systems alike to the one being used in parallel here. For those systems, a predetermined value is used to trigger acquisition of the signal when this is exceeded. Here, the acquisition is constant, with this threshold only being an additional pre-processing phase to limit the effect of noise on the signal, with the selected value being significantly lower and selected in a dynamic manner.

3.1 Application of the Root Mean Square

The Root Mean Square (RMS) gives the average amplitude of the signal over a series of points. This can be calculated through the usage of the following:

$$RMS = \sqrt{\frac{\sum_{n=1}^N V(n)^2}{N}} \quad (1)$$

where N = number of datapoints (7). For the application of RMS to the captured signal, a moving, scanning, approach should be implemented. This involves the

application of a moving uniform window to the data, where the length of the window corresponds to the value of N, where overlap has been introduced between consecutive windows. This is successfully implemented through the usage of existing applications within MATLAB, specifically the dsp.MovingRMS (8). The usage of this available algorithm allows for the completion of the above procedure in a more efficient manner than by implementation of a custom function within MATLAB, hence its preferential usage.

3.2 Investigation of the Effects of Window Length

Assigning the correct window length for the given signal is key to the quality of the information that is ascertainable. It is important to balance the precision of the window length alongside the performance of the system. A variety of selected window lengths can be utilised in useful means, with this being a question of the required level of information.

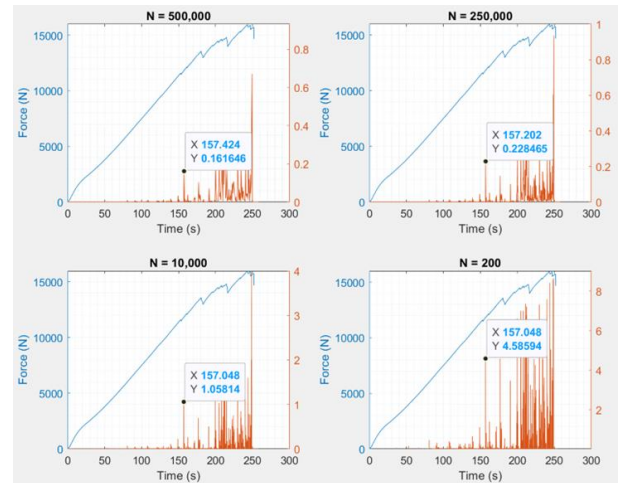


Fig. 1. Comparison of the Window Length on the Produced Representation

Figure 1 demonstrates the effect of varying window length utilising the same AE signal and tensile test data. The test data produced from the tensile test of coupon T02 was used for this example as it most clearly demonstrates the highlighted observable factor. The highlighted peaks within the four representations in the figure corresponds to the first major delamination event. When using a smaller window length, there is the presence of surrounding peaks to the identified delamination event; particularly apparent for window length of 200.

Table 1. Effect of Window Length N the Produced Numerical RMS Representation

N	Number of Datapoints	Reduction Factor	Identified Time of Delamination
5×10^5	473	2.4545×10^5	157.4237
2.5×10^5	989	2.6087×10^5	157.2024
10^4	2.5757×10^4	1.0017×10^4	157.0478
2×10^2	1.296407×10^6	199.0116	157.0475

For the assignment of the correct window length consideration of various factors based on the observations made above is key.

Within the application of the scanning windowing here, the overlap was set to +1. As uniform windows are utilised, there is no need for the introduction of significant overlap for the retention of signal information as is the case with alternate windowing types. An additional benefit of the usage of minimal overlap, is the retention of a true representation of the signal, which given the intended usage, is vital for the correct evaluation and extraction of damage events and their corresponding time vector.

3.3 Assignment of Damage Modes

The accurate and appropriate assignment of damage based on the information provided by this approach is arguably the most important factor in determining its effectiveness for expansion and continued usage.

3.3.1 Intensity Based Approach

One such approach for the assignment of damage modes based on information from AE is the evaluation of the peak amplitude associated with the AE response produced by the occurrence of damage. The approach operates on the principle that specific damage modes can be directly attributed to a set range of amplitude values which the peak falls between. Usage of exact literature values may not be advisable due to different amplification and other such parameters between the systems. This is a fairly simplistic approach to assigning damage, only assessing the peak, or dominant damage mode. Viewing the RMS as an average of the signal, similar principles can be applied when evaluating this approach. However, the peak amplitude cannot be directly assessed as it is encapsulated within the windowed signal. Understanding of specific damage mechanisms can corroborate assigned relative positions of the damage modes.

Another heavily subscribed approach is the assessment of the AE energy, essentially the energy encapsulated by the waveform itself. There are two approaches which can be used for the calculation of this, either the Measure Area under the Rectified Signal Envelope (MARSE) energy or the true energy, area under the squared signal envelope (9). There are inherent issues associated with the usage of either method. For the MARSE approach only the signals within the positive range are accounted for, whereas for the true energy, all points are accounted for, as the signal is squared, however, this will introduce skewing to the values when considering them all in a single representation, e.g. those early in the loading cycle when there are few AE events versus those later in the signal where the signal becomes heavily saturated due to the frequency of AE events and damage at this point in the loading cycle (10,11).

The means by which each of these is used to assess damage varies significantly, however, the reasoning used for how each of these is applied is based around

similar logic. Both of these can be related to the observed energy of the associated damage mechanism, with their position relative to the other modes reflecting roughly the energy for their respective mechanisms. For both methods there is some discussion regarding the precise positioning of some damage modes. However, there appears to be consensus on the positioning of matrix cracking, delamination and fibre fracture, in this order, with fibre-matrix debonding and fibre-pullout in varying positions. For an amplitude-based approach, as there is a set range, precise values are available for comparison. However, the ranges provided demonstrate large variation throughout literature. This shows dependence on a series of variables, primarily material type, hence its relative acoustic impedance, as well as variation in established units used when considering the response (12–15), yet they hold true to the established trends outlined above. A key means of identifying the occurrence of a damage event in composites is through the monitoring of the rate of change associated with the cumulative energy seen for the system. It should be noted that individual AE energy values can be compared within a system if a maximum value of a single peak is known. This allows established trends, similar to what is observed for amplitude-based approach, to be assigned to energy ranges within that given system. These follow the expected positioning, with matrix cracking being the lowest energy, then delamination and finally fibre breakage. Similar observations regarding the difficulty of assigning debonding and pullout to what was discussed previously were seen here (16).

Accounting for the varying methods as well as the established links between these and RMS, a valid direction for evaluation, would be a mixed approach which takes advantage of both the discussed alternatives, applying similar logic as well as utilising the established trends encountered above. It can be viewed in this manner as it has a set range of values, however, the correspondence between the observable RMS values and the damage mode responsible is harder to relate as there is a series of factors which can affect this. These factors are partly what can be seen as variables for the other methods as well and need to be considered when utilising AE. Using a scanning approach, moving RMS, will be similar in what it can discern to the other discussed techniques, in that at a simplistic level it can only accurately give clear indication of the dominant damage mode within the window. The main difference between the technique discussed and others is simply that a true representation of the original signal is maintained throughout.

As the application of this technique is based around the processing of the signal acquired by the customised system, it is optimised for the availability of raw data, however, the commercial systems RMS values do reflect the trends highlighted, such that similar logic may also be applied there. It should be noted that for this and the following sections that the points and assumption made, are done so under the pretext of the conditions provided by the customised system.

3.3.2 Extraction of Corresponding Waveform

Despite the explored similarities between this technique and other methods for the assignment of the dominant damage mode, there is one real advantage to the preferential usage of this technique. That is the inherent ability for the isolation and extraction of any associated damage event as the corresponding AE response is sectioned into the relating windows within processing. Such an approach gives this technique the benefit of being a live constant monitoring approach with direct access to corresponding AE data. This is unlike conventional approaches, where such information requires exporting, with the quality of such data being subject to some debate.

Naturally this approach is not without issue, requiring some further considerations. These issues may affect the value calculated during signal processing, as well as the corresponding waveform. The primary concern can be considered to be the risk of segmentation of a single signal, resulting in crossover between multiple windows, as well as the risk of inclusions within the captured windows. The segmentation of a single waveform is not unique to this technique, and has been observed within different systems, with this essentially being the spanning of a waveform across the boundary of two separate windows. This risk increases as the window length is reduced, however, this is relatively simple to rectify, it requires the extraction of both windows involved. The main concern would be the effect on the calculated RMS values, however, from proper evaluation of the produced representation the occurrence of this can be confirmed through the presence of successive peaks within one window length of each other. Other than this, the risk of noise inclusions may affect the calculated RMS value, the main source of these being from electronic or mechanical sources. Although some precautions have been taken in order to account for this, through the application of a dynamic threshold, as the production of this noise can be expected to increase as the intensity of the mechanical testing also increases. Overall, on windows containing AE events the effect of the presence of this will be minimal on the calculated RMS value. The real predicament is the windows containing only this producing RMS peaks, however, these can simply be accounted for through slight filtering of the produced RMS results if required, as the peaks produced by these windows will be minimal in intensity.

Another occurrence which needs to be accounted for is the presence of precursor or subsequent signals which may either be included within the waveform or be associated with it in some form. As mentioned, this may result in their exclusion from the isolated signal, however, this should in theory only be applicable when smaller window lengths are used. In terms of the effect of this on the evaluation of damage modes, this is dependent not only on the extent of the separation from the signal, but also on the relation, whether they are intrinsically linked or simply just in close proximity to the primary signal. These signals may also be attributed to the different wave types which contribute to the AE signal, and the varying associated parameters of each of

these. The wave type is deterministic in the behaviour and characteristics of the associated AE, such as the wave speed which can be directly attributed to the observed time of arrival. Such a phenomenon is due to the means by which respective waves travel within the material in question, however, the type is not specified by the damage type.

A clear benefit associated with the use of such a method is the availability and ease of application of further signal processing techniques to specified windows highlighted by initial evaluation of the produced RMS representation. This approach can be utilised to confirm whether the assigned damage modes from the RMS representation hold true under scrutiny of more complex and rigorous investigation. A benefit of applying this methodology is simply limiting the amount of data which is used as input for more computationally complex and demanding algorithms are to be applied to. One such further signal processing technique is the application of Fourier Transform (FT), as to allow for evaluation of the associated frequency components.

3.3.3 Confirmation of Damage Mode Using Fourier Transform

Despite the evaluation of amplitude being an established and rather heavily used technique, it does have some inherent limitations and flaws. Primarily it operates under the pretext that damage modes occur independently, whereas in actuality some of these are intrinsically linked. There is also the possibility of such events occurring simultaneously or at similar times at different locations. For appreciation of these more complex, interlinked, events expansion of the signal processing operations to include a more comprehensive technique is required. Final confirmation of the conclusions made from previous evaluations can be completed through the application of FT to the extracted signal envelope of the corresponding RMS window. This technique evaluates the frequency components of an inputted signal, with each distinctive damage mode having a characteristic range of associated frequencies, which in turn allows for a full appreciation of damage modes occurring within a given signal.

Adhering to the Nyquist theorem (17), the maximum frequency that can be accurately sampled within the established methodology is 500 kHz, half of the sample rate of 1 MS/s (MHz). This is done purposely as to fit with the established frequencies which are associated with the occurrence of damage within composite materials. Sampling below this rate would result in the omission of higher frequencies, sampling at a higher rate than this, can be beneficial, however, on the whole these can be viewed as creating what can be viewed as being excessive amounts of data, which are not necessary for accurate analysis.

Likewise to the amplitude approach considered prior, the assignment of frequency ranges to damage is a point of contention and debate. However, similar logic is used for assigning the frequency ranges to damage modes, as the frequency of the associated wave can be directly related to the energy of the source, where the higher the frequency, the greater the associated energy.

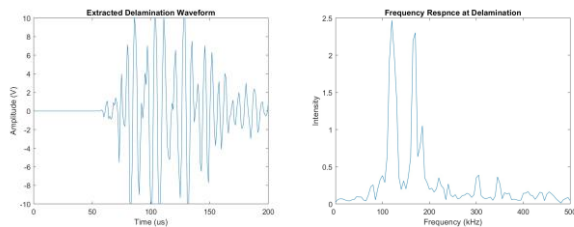


Fig 2. Extracted Wavelength and Corresponding Fourier Transform for Identified Major Delamination Event

Specifically in this case the discrete form of FT was selected, due to the set time length of the window. These DFT were completed using the inbuilt algorithm in MATLAB, utilising the specified procedure for the application of the technique to noisy signals (18,19). Frequency based analysis of damage tends to be variable in a similar manner as to what was explored for the other signal processing methods, in that defining specific values or ranges attributed to damage modes is a point of contention. The two clear peaks in the above representation can be assigned to matrix cracking and delamination, with the latter being the higher intensity of the two. This fits with what is seen within the stress curve, and acts as further confirmation for the conclusions drawn from the RMS approach.

4 Conclusions

The moving RMS approach to the evaluation of AE data can provide an alternative to pre-established techniques for the assessment of damage within FRPs. It has been outlined that they can be used to determine the same information, at the same level of detail and accuracy as these. This is achieved whilst maintaining a true representation of the data whilst allowing for the extraction of raw associated waveform which allow for further signal processing techniques to be applied to the data within the associated window. This can be viewed as the real benefit to the preferential selection of this technique in conjunction with the customised AE acquisition system. The outlined approach here can be viewed as initial stages in the development of a true quantitative approach, giving a much more in-depth assessment of the total damage observed for the system, whilst still providing means for simple evaluation of dominant damage modes. It should also be noted that the RMS information provided by the commercial system also agrees with the conclusions drawn here providing accessibility to a simpler version of the discussed technique. Such that it can be used in conjunction with the other available values provided in order to confirm the assignment of damage modes.

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